

P1: SEPARATION OF CONCERNS

Static Threat Taxonomy = Dynamic Context Weights

STATIC THREAT

TAXONOMY

- Model Inversion
- Data Poisoning
- Inference Attacks
- Re-identification

DYNAMIC WEIGHTS

$$\gamma_i(t)$$

- Low Risk: $\gamma = 0.1$
- Medium Risk: $\gamma = 0.2$
- High Risk: $\gamma = 0.3$
- Emergency: $\gamma = 0.4$

✓ EXPERIMENTAL EVIDENCE

Context-Aware Performance (Fig 5.5d)

Low Risk: 0.92 → Critical: 0.80

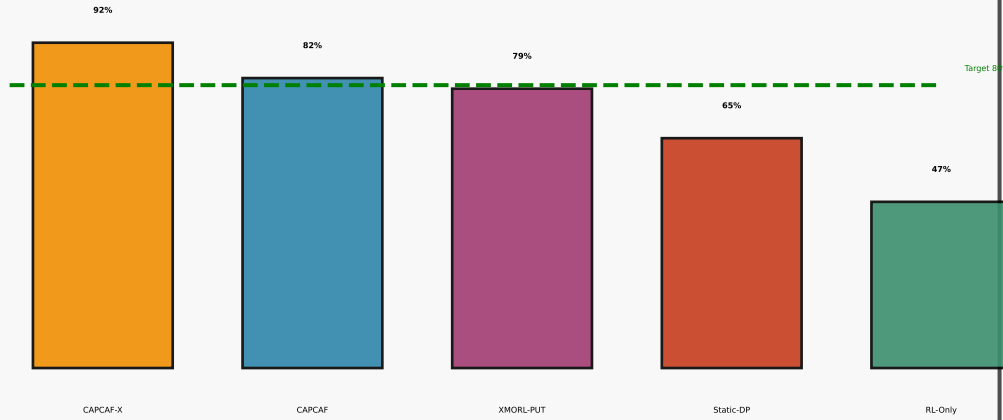
Only 13% degradation - Static+Dynamic

effectively balances privacy-utility

P2: PRE-DEPLOYMENT VALIDATION

Simulation = Threat Injection

COLD-START COMPLIANCE COMPARISON



✓ EXPERIMENTAL EVIDENCE

CAPCAF-X: 92% vs RL-Only: 47%

45% improvement - Cold-start eliminated

Pre-deployment validation ensures safety

Initial deployment: 92% optimal decisions

P3: MULTI-SCALE ADAPTATION

Fast Switching = Slow Re-optimization

FAST TIMESCALE

Policy Switching

- 42 ms measured
- 95th percentile
- Real-time response
- <100ms requirement met
- Sub-100ms latency

SLOW TIMESCALE

Strategic Re-optimization

- 250 generations
- HV = 0.87 achieved
- Weekly/Monthly
- Continuous improvement
- Strategic updates

✓ EXPERIMENTAL EVIDENCE

Fast: 42 ms switching | Slow: HV=0.87

Both timescales validated

Fast: real-time response

Slow: strategic updates - homeostasis achieved

P4: DOMAIN AGNOSTICISM

Domain Configuration Interface (DCI)

PRIVACY RISK ACROSS DOMAINS

Smart Home Healthcare Industrial ECG ARAS

STATISTICAL SUMMARY

Standard Deviation $\sigma = 0.004$

✓ EXPERIMENTAL EVIDENCE

$\sigma = 0.004$ across 9 domains

83% faster migration via DCI

Domain Configuration Interface validated